Aalto University

# Moment Estimator-Based Extreme Quantile Estimation with Erroneous Observations

Joint work with Pauliina Ilmonen and Lauri Viitasaari
https://doi.org/10.48550/arXiv.2502.08510

Jaakko Pere

Thursday 28th May, 2026

# Agenda of the Presentation

Motivation

Preliminaries of Extreme Value Theory

Extreme Value Theory with Approximations

Extreme Quantile Region Estimation under Ellipticity

# Table of Contents

## Extreme Value Theory: Tools for Extrapolation

- Extreme value theory is concerned with inference of rare events.

# Extreme Value Theory: Tools for Extrapolation

- Extreme value theory is concerned with inference of rare events.
- Consider estimating 0.999-quantile (red cross) with a reasonable sample (black circles).

# Extreme Value Theory: Tools for Extrapolation

- Extreme value theory is concerned with inference of rare events.
- Consider estimating 0.999-quantile (red cross) with a reasonable sample (black circles).

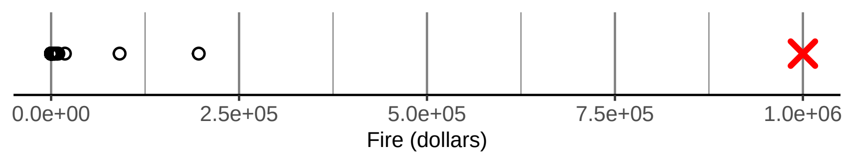



**Figure:** Insurance company keeps records of fire insurance claims per windstorm. In total, the figure shows 700 observations.

## Noisy Observations

What if instead of the original observations only approximations are available?

$\Rightarrow$ Extreme value inference with the noisy sample.

# Noisy Observations

What if instead of the original observations only approximations are available?

$\Rightarrow$ Extreme value inference with the noisy sample.

Why?

## Noisy Observations

What if instead of the original observations only approximations are available?

$\Rightarrow$ Extreme value inference with the noisy sample.

Why?

- Extreme value index estimation for projections of functional data (Kokoszka and Xiong 2018).
- Extreme quantile/expectile regression (Girard et al. 2021).
- Latent model extreme value index estimation (Virta et al. 2024).

## Noisy Observations

What if instead of the original observations only approximations are available?

$\Rightarrow$ Extreme value inference with the noisy sample.

Why?

- Extreme value index estimation for projections of functional data (Kokoszka and Xiong 2018).
- Extreme quantile/expectile regression (Girard et al. 2021).
- Latent model extreme value index estimation (Virta et al. 2024).

Much more work has been done especially in the heavy-tailed case (few or no moments exist)...

## Approaches from the Literature (for the Heavy Tail)

1. (Kim and Kokoszka 2020; Kim and Kokoszka 2023): We have $\hat{Y}_i = Y_i + \varepsilon_i$, where

- the errors $\varepsilon_i$ are i.i.d.,
- $\{Y_i\} \perp\!\!\!\perp \{\varepsilon_i\}$, and
- $\lim_{t\to\infty} \mathbb{P}(|\varepsilon| > t)/\mathbb{P}(X_1 > t) = 0$

# Approaches from the Literature (for the Heavy Tail)

1. (Kim and Kokoszka 2020; Kim and Kokoszka 2023): We have $\hat{Y}_i = Y_i + \varepsilon_i$, where
   - the errors $\varepsilon_i$ are i.i.d.,
   - $\{Y_i\} \perp\!\!\!\perp \{\varepsilon_i\}$, and
   - $\lim_{t\to\infty} \mathbb{P}(|\varepsilon| > t)/\mathbb{P}(X_1 > t) = 0$
2. (Girard et al. 2021): We have that the errors $|\hat{Y}_i - Y_i|$
   - can be dependent,
   - they do not need to be identically, distributed, and
   - $\{Y_i\} \perp\!\!\!\perp \{\varepsilon_i\}$ can fail.
   - "Quantiles of $|\hat{Y}_i - Y_i|$ grow sufficiently slowly"

## Common message in the heavy-tailed case

If errors are less heavy-tailed compared to the observations, life is good.

## Scope of the Presentation

1. We quantify the effect of approximation errors for a certain *extreme quantile estimator*:
   - Short-, light-, and heavy-tailed distributions are considered.

## Scope of the Presentation

1. We quantify the effect of approximation errors for a certain *extreme quantile estimator*:
   - Short-, light-, and heavy-tailed distributions are considered.
2. Application to multivariate extreme quantile region estimation under ellipticity.

# Table of Contents

## Maximum Domain of Attraction ($Y \in \mathrm{MDA}(G)$)

Let $Y_1, \ldots, Y_n$ be i.i.d. observations of a random variable $Y$. We have $Y \in \mathrm{MDA}(G)$ if there exist sequences $a_n > 0$ and $b_n \in \mathbb{R}$, and a random variable $G$ with a nondegenerate distribution such that

$$\frac{\max(Y_1, \ldots, Y_n) - b_n}{a_n} \xrightarrow{\mathcal{D}} G, \quad n \to \infty.$$

## Extreme Value Index (Fisher and Tippett 1928; Gnedenko 1943)

Up to location and scale, the distribution of $G = G_\gamma$ is characterized by the parameter $\gamma$, called the *extreme value index*. That is, the distribution of $G_\gamma$ is of the type

$$F_{G_\gamma}(x) = \begin{cases} \exp\left(-\left(1+\gamma x\right)^{-1/\gamma}\right), & 1+\gamma x > 0 \quad \text{if} \quad \gamma \neq 0, \\ \exp\left(-e^{-x}\right), \quad x \in \mathbb{R} & \text{if} \quad \gamma = 0. \end{cases}$$

Maximum domain of attraction $\mathrm{MDA}\,(G_\gamma)$ for different values of the extreme value index $\gamma$.

| $\gamma$ | Example of $F \in \mathrm{MDA}\,(G_\gamma)$ | Properties of $F$ |
|---|---|---|
| $> 0$ | Pareto distribution | For a larger $\gamma$ fewer moments exist* |
| $= 0$ | Normal distribution | All moments exist* |
| $< 0$ | Uniform distribution | Finite endpoint |

*The claims about the moments hold if the left tail of a distribution is not considered.

## Tail Quantile Function

Define the tail quantile function corresponding to a distribution $F$ by

$$U(t) = F^{\leftarrow}\left(1 - \frac{1}{t}\right), \quad t > 1,$$

where we denote the left-continuous inverse of a nondecreasing function by $f^{\leftarrow}(y) = \inf\{x \in \mathbb{R} : f(x) \geq y\}$.

## Tail Quantile Function

Define the tail quantile function corresponding to a distribution $F$ by

$$U(t) = F^{\leftarrow}\left(1 - \frac{1}{t}\right), \quad t > 1,$$

where we denote the left-continuous inverse of a nondecreasing function by $f^{\leftarrow}(y) = \inf\{x \in \mathbb{R} : f(x) \geq y\}$.

That is, $U(1/p)$ is the $(1-p)$-quantile.

## MDA Condition in Terms of Quantiles (de Haan 1984)

Let $U$ be the tail quantile function corresponding to a random variable $Y$. Then, $Y \in \mathrm{MDA}\,(G_\gamma)$ if and only if there exists a positive function $a : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x > 0$,

$$\lim_{t \to \infty} \frac{U\,(tx) - U\,(t)}{a\,(t)} = \frac{x^\gamma - 1}{\gamma}.$$

## MDA Condition in Terms of Quantiles (de Haan 1984)

Let $U$ be the tail quantile function corresponding to a random variable $Y$. Then, $Y \in \mathrm{MDA}\,(G_\gamma)$ if and only if there exists a positive function $a : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x > 0$,

$$\lim_{t\to\infty} \frac{U\,(tx) - U\,(t)}{a\,(t)} = \frac{x^\gamma - 1}{\gamma}.$$

Choose $t = n/k$ and $x = k/(np)$ to get the approximation

$$U\left(\frac{1}{p}\right) \approx U\left(\frac{n}{k}\right) + a\left(\frac{n}{k}\right)\frac{\left(\frac{k}{np}\right)^\gamma - 1}{\gamma}.$$

## Extreme Quantile Estimation

Suppose $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$ is an i.i.d. sample of $Y \in \mathrm{MDA}\,(G_\gamma)$, $\gamma \in \mathbb{R}$. Denote order statistics corresponding to the sample $\boldsymbol{Y}$ by $\boldsymbol{Y}_{1,n} \leq \cdots \leq \boldsymbol{Y}_{n,n}$. Then an estimator for the extreme $(1-p)$-quantile $x_p = U\,(1/p)$ can be given as

$$\hat{x}_p\,(\boldsymbol{Y}) = \boldsymbol{Y}_{n-k,n} + \hat{a}\left(\frac{n}{k}\right) \frac{\left(\frac{k}{np}\right)^{\hat{\gamma}(\boldsymbol{Y})} - 1}{\hat{\gamma}\,(\boldsymbol{Y})},$$

where $\hat{\gamma}$ is an estimator for the extreme value index $\gamma$.

## The Moment Estimator (Dekkers et al. 1989)

Set, for $\ell \in \{1, 2\}$,

$$M_n^{(\ell)}(\boldsymbol{Y}) = \frac{1}{k} \sum_{j=0}^{k-1} \left(\ln \boldsymbol{Y}_{n-j,n} - \ln \boldsymbol{Y}_{n-k,n}\right)^{\ell},$$

and

$$\hat{\gamma}_{+}(\boldsymbol{Y}) = M_n^{(1)}(\boldsymbol{Y}) \quad \text{and} \quad \hat{\gamma}_{-}(\boldsymbol{Y}) = 1 - \frac{1}{2} \left(1 - \frac{\left(M_n^{(1)}(\boldsymbol{Y})\right)^2}{M_n^{(2)}(\boldsymbol{Y})}\right)^{-1}.$$

Then the moment estimator is defined as

$$\hat{\gamma}_M(\boldsymbol{Y}) = \hat{\gamma}_{+}(\boldsymbol{Y}) + \hat{\gamma}_{-}(\boldsymbol{Y}),$$

and the corresponding estimator for $a(n/k)$ is defined as

$$\hat{\sigma}_M(\boldsymbol{Y}) = \boldsymbol{Y}_{n-k,n} M_n^{(1)}(\boldsymbol{Y}) \left(1 - \hat{\gamma}_{-}(\boldsymbol{Y})\right).$$

# Table of Contents

## Back to Approximations

- What if instead of the sample $\boldsymbol{Y}$, only approximations $\hat{\boldsymbol{Y}} = (\hat{Y}_1, \ldots, \hat{Y}_n)$ are available?

## Back to Approximations

- What if instead of the sample $\boldsymbol{Y}$, only approximations $\hat{\boldsymbol{Y}} = (\hat{Y}_1, \ldots, \hat{Y}_n)$ are available?
- How the approximation error affects the asymptotics?

## Back to Approximations

- What if instead of the sample $\boldsymbol{Y}$, only approximations $\hat{\boldsymbol{Y}} = (\hat{Y}_1, \ldots, \hat{Y}_n)$ are available?
- How the approximation error affects the asymptotics?

That is, next we quantify how the error

$$\max_{0 \le j \le k} \left| \frac{\hat{\boldsymbol{Y}}_{n-j,n}}{\boldsymbol{Y}_{n-j,n}} - 1 \right|$$

has to behave in order to have asymptotic normality for $\hat{x}_p(\hat{\boldsymbol{Y}})$.

## Assumptions for the Main Result

1. $Y$ is an almost surely positive random variable with $F \in \mathcal{D}\left(G_\gamma\right)$, $\gamma \in \mathbb{R}$ (and $U$ satisfies certain second-order conditions).
2. $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$ are i.i.d. copies of $Y$ and let $\hat{\boldsymbol{Y}} = (\hat{Y}_1, \ldots, \hat{Y}_n)$ be an approximated sample.

## Assumptions for the Main Result

1. $Y$ is an almost surely positive random variable with $F \in \mathcal{D}(G_\gamma)$, $\gamma \in \mathbb{R}$ (and $U$ satisfies certain second-order conditions).
2. $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$ are i.i.d. copies of $Y$ and let $\hat{\boldsymbol{Y}} = (\hat{Y}_1, \ldots, \hat{Y}_n)$ be an approximated sample.
3. Suppose $k = k_n \to \infty, k_n/n \to 0$, as $n \to \infty$.
4. $p = p_n$, $np_n = o(k_n)$ and $\ln(np_n) = o\left(\sqrt{k_n}\right)$.

## Sketch of the Main Result

Suppose that assumptions of the previous slide hold and that

$$\max_{0\le j\le k_n}\left|\frac{\hat{\boldsymbol{Y}}_{n-j,n}}{\boldsymbol{Y}_{n-j,n}}-1\right|=O_{\mathbb{P}}\left(h_n\right)$$

for some sequence $h_n$ such that $\sqrt{k_n}\frac{h_n U\left(\frac{n}{k_n}\right)}{a\left(\frac{n}{k_n}\right)}=o(1)$. Denote $q_\gamma\left(t\right)=\int_1^t s^{\gamma-1}\ln s\,\mathrm{d}s$. Then

$$\sqrt{k_n}\frac{\hat{x}_{p_n}(\hat{\boldsymbol{Y}})-U\left(1/p_n\right)}{a\left(\frac{n}{k_n}\right)q_\gamma\left(\frac{k_n}{np_n}\right)}$$

is asymptotically normal.

## Condition for the Error

Suppose $k = k_n \to \infty, k_n/n \to 0$, as $n \to \infty$. Assume also that $U$ satisfies a certain second-order condition and that

$$\max_{0 \le j \le k_n} \left| \frac{\hat{\boldsymbol{Y}}_{n-j,n}}{\boldsymbol{Y}_{n-j,n}} - 1 \right| = O_{\mathbb{P}}(h_n)$$

for some sequence $h_n$. Then $\sqrt{k_n} \frac{h_n U\left(\frac{n}{k_n}\right)}{a\left(\frac{n}{k_n}\right)} \to 0$ if

1. $\gamma > 0$ and $\sqrt{k_n} h_n \to 0$,
2. $\gamma = 0$ and $\sqrt{k_n} h_n = O\left(\left(\frac{n}{k_n}\right)^{-\delta}\right)$ for some $\delta > 0$,
3. $\gamma < 0$ and $\sqrt{k_n} h_n = o\left(\left(\frac{n}{k_n}\right)^{\gamma}\right)$.

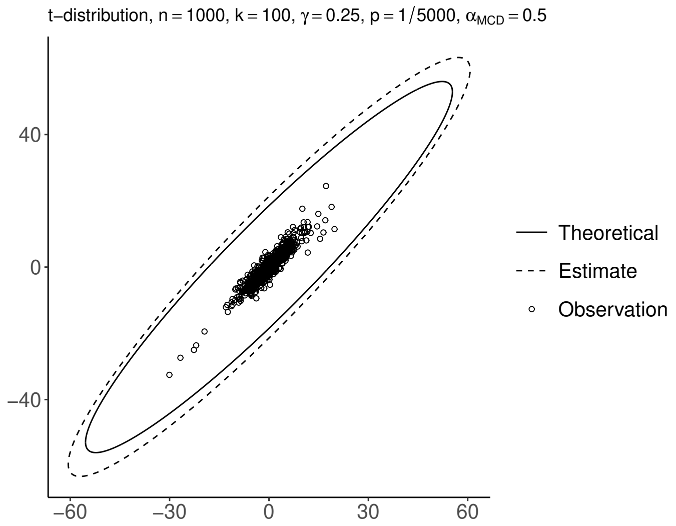
## Example

If

$$\max_{0\le j\le k_n}\left|\frac{\hat{\boldsymbol{Y}}_{n-j,n}}{\boldsymbol{Y}_{n-j,n}}-1\right|=O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right),$$

then the sufficient condition for the error does not hold if $\gamma \le -1/2$!

## Table of Contents


t−distribution, n = 1000, k = 100, γ = 0.25, p = 1/5000, $\alpha_{MCD}$ = 0.5
40
0
−40
−60
−30
0
30
60
Theoretical
Estimate
Observation

## Extreme Quantile Region Estimation

Let $X \in \mathbb{R}^d$ be a random vector with density $f$. We wish to estimate

$$Q_p = \left\{ x \in \mathbb{R}^d : f(x) \leq \beta_p \right\},$$

where $\beta_p$ is chosen such that $\mathbb{P}(X \in Q_p) = p$, and $p$ is small.

## Extreme Quantile Region Estimation

Let $X \in \mathbb{R}^d$ be a random vector with density $f$. We wish to estimate

$$Q_p = \left\{ x \in \mathbb{R}^d : f(x) \leq \beta_p \right\},$$

where $\beta_p$ is chosen such that $\mathbb{P}(X \in Q_p) = p$, and $p$ is small.

The problem has been considered under multivariate regular variation by

- Cai et al. 2011 in the context of density contours, and
- He and Einmahl 2017 in the context of halfspace depth contours

under multivariate regular variation.

## Elliptical Distributions (Cambanis et al. 1981)

We say that a $d$-variate random vector $X$ is elliptically distributed if

$$X \overset{\mathcal{D}}{=} \mu + \mathcal{R}\Lambda S,$$

where

- $\mu \in \mathbb{R}^d$ is called the *location vector*,
- $\mathcal{R}$ is a nonnegative random variable called the *generating variate*,
- $\Lambda \in \mathbb{R}^{d \times d}$ is a matrix such that $\Sigma = \Lambda\Lambda^\mathsf{T}$ is a symmetric positive definite matrix (matrix $\Sigma$ is called *scatter matrix*),
- $S$ is an $d$-dimensional random vector uniformly distributed over the unit-sphere $\{x \in \mathbb{R}^d : x^\mathsf{T} x = 1\}$ and
- Random variables $\mathcal{R}$ and $S$ are independent.

## Role of the Generating Variate $\mathcal{R}$

- We suppose $\mathcal{R} \in \mathrm{MDA}\,(G_\gamma)$ for $\gamma \in \mathbb{R}$

## Role of the Generating Variate $\mathcal{R}$

- We suppose $\mathcal{R} \in \text{MDA}\,(G_\gamma)$ for $\gamma \in \mathbb{R}$
- $\mathcal{R} \stackrel{\mathcal{D}}{=} \sqrt{(X-\mu)^\mathsf{T}\,\Sigma^{-1}\,(X-\mu)}$

## Role of the Generating Variate $\mathcal{R}$

- We suppose $\mathcal{R} \in \mathrm{MDA}\,(G_\gamma)$ for $\gamma \in \mathbb{R}$
- $\mathcal{R} \stackrel{\mathcal{D}}{=} \sqrt{(X - \mu)^\intercal \, \Sigma^{-1} \, (X - \mu)}$
- $\hat{R}_i = \sqrt{(X_i - \hat{\mu}(\boldsymbol{X}))^\intercal \, \hat{\Sigma}^{-1}(\boldsymbol{X}) \, (X_i - \hat{\mu}(\boldsymbol{X}))}$
- $\hat{\boldsymbol{R}} = (\hat{R}_1, \ldots, \hat{R}_n)$

## The Estimator

- We wish to estimate

$$Q_p = \left\{ x \in \mathbb{R}^d : \sqrt{(x-\mu)^\mathsf{T}\, \Sigma^{-1}\, (x-\mu)} \geq r_p \right\},$$

for a very small $p$, where $r_p$ is the $(1-p)$-quantile of the generating variate $\mathcal{R}$.

## The Estimator

- We wish to estimate

$$Q_p = \left\{ x \in \mathbb{R}^d : \sqrt{(x-\mu)^\mathsf{T}\, \Sigma^{-1}\, (x-\mu)} \geq r_p \right\},$$

for a very small $p$, where $r_p$ is the $(1-p)$-quantile of the generating variate $\mathcal{R}$.

- Define estimator $\hat{Q}_p$ as

$$\hat{Q}_p = \left\{ x \in \mathbb{R}^d : \sqrt{(x-\hat{\mu}(\boldsymbol{X}))^\mathsf{T}\, \hat{\Sigma}^{-1}(\boldsymbol{X})\, (x-\hat{\mu}(\boldsymbol{X}))} \geq \hat{x}_p(\hat{\boldsymbol{R}}) \right\},$$

where $\hat{x}_p(\hat{\boldsymbol{R}})$ is the extreme quantile estimator computed with approximations $\hat{\boldsymbol{R}}$.

## Approximation Error for the Generating Variate

Assume $k = k_n \to \infty, k_n/n \to 0$, as $n \to \infty$ and that estimators of the location and scatter satisfy $\sqrt{n}\left(\hat{\mu}_n - \mu\right) = O_{\mathbb{P}}\left(1\right)$ and $\sqrt{n}\left(\hat{\Sigma}_n - \Sigma\right) = O_{\mathbb{P}}\left(1\right)$. Then

$$\max_{0 \leq j \leq k_n} \left| \frac{\hat{\boldsymbol{R}}_{n-j,n}}{\boldsymbol{R}_{n-j,n}} - 1 \right| = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right).$$

# Consistency and Affine Equivariance

- **Consistency:** if estimators of the location and scatter are $\sqrt{n}$-consistent, $\mathcal{R} \in \mathrm{MDA}\,(G_\gamma)$ for $\gamma > -1/2$, $p_n \to 0$ with a suitable rate, as $n \to \infty$ (and other technical conditions),

$$\frac{\mathbb{P}\left(X \in \hat{Q}_{p_n} \triangle Q_{p_n}\right)}{p_n} \to 0, \quad \text{as } n \to \infty,$$

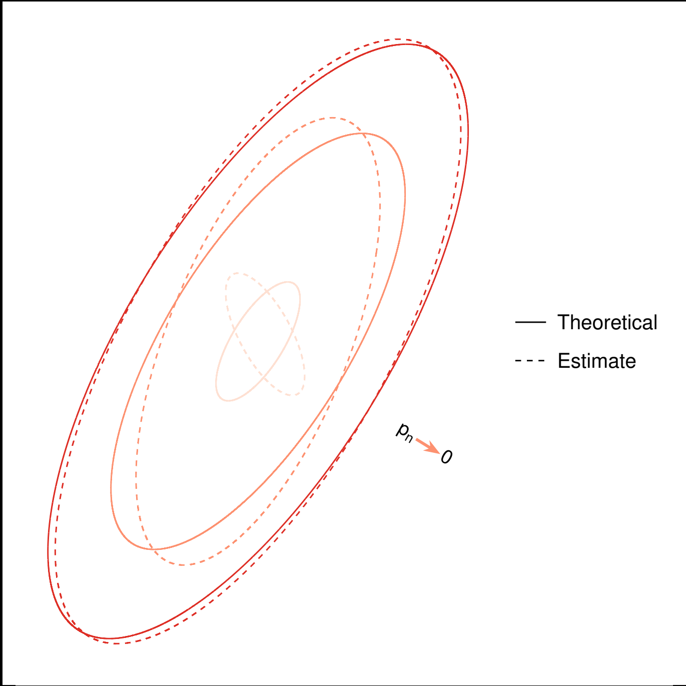
where $B_1 \triangle B_2 = (B_1 \setminus B_2) \cup (B_2 \setminus B_1), \quad B_1, B_2 \in \mathbb{R}^d$.

## Consistency and Affine Equivariance

- **Consistency:** if estimators of the location and scatter are $\sqrt{n}$-consistent, $\mathcal{R} \in \mathrm{MDA}\,(G_\gamma)$ for $\gamma > -1/2$, $p_n \to 0$ with a suitable rate, as $n \to \infty$ (and other technical conditions),

$$\frac{\mathbb{P}\left(X \in \hat{Q}_{p_n} \triangle Q_{p_n}\right)}{p_n} \to 0, \quad \text{as } n \to \infty,$$

  where $B_1 \triangle B_2 = (B_1 \setminus B_2) \cup (B_2 \setminus B_1), \quad B_1, B_2 \in \mathbb{R}^d$.
- **Equivariance under affine transformations** $Ax + b$, $A \in \mathbb{R}^{d \times d}$ invertible.



Theoretical
Estimate
$p_n \to 0$

Link to slides: `https://github.com/perej1/presentations/tree/main/light-error`

## References

Cai, Juan-Juan, John H. J. Einmahl, and Laurens de Haan (2011). "Estimation of extreme risk regions under multivariate regular variation". In: *The Annals of Statistics* 39.3, pp. 1803–1826. DOI: 10.1214/11-AOS891.

Cambanis, S., S. Huang, and G. Simons (1981). "On the theory of elliptically contoured distributions". In: *Journal of Multivariate Analysis* 11.3, pp. 368–385.

de Haan, Laurens (1984). "Slow Variation and Characterization of Domains of Attraction". In: *Statistical Extremes and Applications*. Ed. by J. Tiago de Oliveira. Dordrecht: Springer, pp. 31–48. DOI: 10.1007/978-94-017-3069-3_4.

## References II

Dekkers, A. L. M., J. H. J. Einmahl, and L. de Haan (1989). "A moment estimator for the index of an extreme-value distribution". In: *The Annals of Statistics* 17.4, pp. 1833–1855. URL: `https://www.jstor.org/stable/2241667`.

Fisher, Ronald Aylmer and Leonard Henry Caleb Tippett (1928). "Limiting forms of the frequency distribution of the largest or smallest member of a sample". In: *Mathematical Proceedings of the Cambridge Philosophical Society* 24.2, pp. 180–190. DOI: `10.1017/S0305004100015681`.

Girard, Stéphane, Gilles Stupfler, and Antoine Usseglio-Carleve (2021). "Extreme conditional expectile estimation in heavy-tailed heteroscedastic regression models". In: *The Annals of Statistics* 49.6, pp. 3358–3382. DOI: `10.1214/21-AOS2087`.

## References III


Gnedenko, Boris (1943). "Sur La Distribution Limite Du Terme Maximum D'Une Série Aléatoire". In: *Annals of Mathematics* 44.3, pp. 423–453. DOI: 10.2307/1968974.

He, Yi and John H. J. Einmahl (2017). "Estimation of extreme depth-based quantile regions". In: *Journal of the Royal Statistical Society. Series B (Statistical Methodology)* 79.2, pp. 449–461. URL: http://www.jstor.org/stable/44682520.

Kim, Mihyun and Piotr Kokoszka (2020). "Consistency of the Hill estimator for time series observed with measurement errors". In: *Journal of Time Series Analysis* 41.3, pp. 421–435. DOI: 10.1111/jtsa.12515.

— (2023). "Asymptotic and finite sample properties of Hill-type estimators in the presence of errors in observations". In: *Journal of Nonparametric Statistics* 35.1, pp. 1–18. DOI: 10.1080/10485252.2022.2136662.

## References IV


Kokoszka, Piotr and Qian Xiong (2018). “Extremes of projections of functional time series on data-driven basis systems”. In: *Extremes* 21, pp. 177–204. DOI: `10.1007/s10687-017-0302-8`.

Virta, Joni et al. (2024). “Latent model extreme value index estimation”. In: *Journal of Multivariate Analysis* 202, p. 105300. DOI: `10.1016/j.jmva.2024.105300`.